Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.2026.DOI

Tri-Level Hybrid DDQN-ALNS: A Hierarchical Learning-Augmented Metaheuristic for the Vehicle Routing Problem with Time Windows

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This research is supported by Ton Duc Thang University.

ABSTRACT The Vehicle Routing Problem with Time Windows (VRPTW) is commonly solved under a lexicographic objective: minimize the number of vehicles (N_{V}) first, then total travel distance (D_{T}). Adaptive Large Neighborhood Search (ALNS) is an established metaheuristic for this problem, and recent work adds learned control to such search. This paper presents Tri-Level Hybrid DDQN-ALNS, which adds three online-learning controllers to an ALNS core: a macro plateau controller modeled approximately as a semi-Markov decision process over aggregated features, a micro operator controller that blends a Dueling Double Deep Q-Network with operator priors through an entropy gate, and a Learned Acceptance Criterion that can accept moves Simulated Annealing rejects. The controllers start from random weights in every run. The search also uses ejection chains, set partitioning with HiGHS, and a frozen Graph Neural Network edge heatmap. We claim neither a new search principle nor a convergence guarantee. We evaluate the method under equal wall-clock budgets of 10, 60, 300 and 900 s on 74 Solomon and Gehring-Homberger instances with five seeds per instance, against ALNS-Base, PyVRP-ILS and an adapted NeuroLKH, on the same machine and in the same blocks, counting only incumbents certified in the original problem space before the cutoff. A factorial ablation on 18 instances separates learning from heuristic components. **[PENDING: two sentences with the rerun's 300 s result against each comparator and the ablation result, including any loss.]**

INDEX TERMS Adaptive Large Neighborhood Search, Deep Reinforcement Learning, Double DQN, Semi-Markov Decision Process, Metaheuristics, Vehicle Routing Problem with Time Windows.

I. INTRODUCTION

THE Vehicle Routing Problem with Time Windows (VRPTW) is an NP-hard combinatorial optimization problem that arises in freight logistics and last-mile parcel delivery [1], [2]. Each route must respect vehicle capacity and the time window of every customer it visits.

Commercial freight planning often treats the objective as lexicographic: dispatchers prioritize minimizing the total number of deployed vehicles (N_{V}) before minimizing total travel distance (D_{T}) [3], [4]. Formally, letting $\mathcal{S}$ denote the space of feasible routing solutions, a solution $s_1 \in \mathcal{S}$ is strictly preferred over $s_2 \in \mathcal{S}$ (denoted $s_1 \prec s_2$) if and only if:

$$N_{\mathrm{V}}(s_1) < N_{\mathrm{V}}(s_2) \ \vee \ \big(N_{\mathrm{V}}(s_1) = N_{\mathrm{V}}(s_2) \wedge D_{\mathrm{T}}(s_1) < D_{\mathrm{T}}(s_2)\big) \tag{1}$$

This ordering reflects the cost structure of commercial transport, where vehicle acquisition, maintenance, insurance and driver wages outweigh marginal fuel cost.

Adaptive Large Neighborhood Search (ALNS) [5], [6] is a leading metaheuristic for VRPTW. It repeatedly destroys and repairs parts of a solution, and its standard form keeps solutions feasible. Standard implementations choose operators with roulette-wheel or bandit rules that aggregate reward

scores over coarse segments. This design has two weaknesses. Operator selection ignores the current search state, such as route fragmentation, temporal slack or capacity use. The search also has no multi-step coordination for leaving plateaus on which removing a vehicle means passing through worse intermediate solutions.

Neural Combinatorial Optimization (NCO) takes a different approach, using Deep Reinforcement Learning (DRL) models such as Pointer Networks [7], Attention Models [8], and constructive policies [9]. These models capture spatial regularities without hand-built heuristics, but end-to-end neural solvers have three problems on VRPTW:

(i) Tight time windows, under which autoregressive decoding can deadlock or need careful penalty tuning [10], [11];
(ii) The objective: they optimize scalarized distance rather than fleet size first ($N_{\mathrm{V}} \succ D_{\mathrm{T}}$); and
(iii) Distribution shift: results often degrade on larger instances or different customer distributions without retuning.

This paper presents Tri-Level Hybrid DDQN-ALNS, which places state-conditioned reinforcement learning inside the search loop of ALNS. It keeps the neighborhood operators of ALNS instead of replacing them with a neural solver, and it returns only solutions validated as feasible. The search has three tiers: a learned macro controller for plateaus, a learned micro controller for operator selection, and a heuristic search tier. Whether this control pays for the compute it consumes is an empirical question, and we treat it as one.

The primary contributions of this paper are summarized as follows:

1) We specify a control layer for ALNS in which a macro plateau controller, a micro operator controller and an acceptance model learn online from random weights in every run, with no offline policy training. Destroy-repair search, Simulated Annealing, ejection chains and set partitioning are established methods, and we do not claim a new search principle. What we study is whether such cold-start control pays for its compute within a time budget.
2) We evaluate the method under equal wall-clock budgets of 10, 60, 300 and 900 s on 74 benchmark instances (56 Solomon-100, 12 Homberger-200, and 6 Homberger-400) with five seeds, against ALNS-Base, PyVRP-ILS and an adapted NeuroLKH. All cases of an instance, seed and cutoff run as one block, and only incumbents certified in the original problem space before the cutoff count.
3) We separate learning from heuristic machinery with a wall-clock factorial ablation over six components at 60 and 300 s, including a 2×2 contrast of the learning and heuristic bundles.
4) We report the limits of the method and of the evidence: where it does not help, how results depend on the budget, the method's earlier dependence on reference fleet sizes, the machine-speed variation we measured, and the way test results influenced development.

The remainder of this paper is organized as follows: Section II reviews related literature. Section III presents the problem formulation. Section IV describes the method. Section V reports the benchmark results, statistical analysis and ablation. Section VI discusses where the method helps, its analytical properties, and its limitations. Finally, Section VII concludes the paper.

II. RELATED WORK

A. CLASSICAL METAHEURISTICS AND ALNS FOR VRPTW

Metaheuristics have long dominated practical VRPTW optimization [4]. Building on Solomon's insertion heuristics [2] and Shaw's Large Neighborhood Search [16], Ropke and Pisinger [5], [6], [17] formalized Adaptive Large Neighborhood Search (ALNS), in which removal and insertion operators compete for selection via roulette-wheel probabilities updated from historical performance. For fleet minimization, Guided Ejection Search [18] and penalty-based edge assembly memetic algorithms [19] target fleet size directly, while matheuristic hybrids periodically solve Set Partitioning models over pools of elite routes [20]–[22]. More recently, Slack Induction by String Removals [12] demonstrated that specialized ruin operators can rival more complex methods.

In standard ALNS, operator weights are exponentially smoothed scores updated over coarse iteration segments, so credit arrives late and in aggregate. The selection policy is stateless: it ignores route topology, cluster fragmentation and time-window tightness. Turkeš et al. [23] found that this adaptive mechanism yields only marginal gains over non-adaptive LNS. That result leaves open whether a better selection policy would help or whether operator selection matters little, which is one reason we measure it directly.

B. NEURAL COMBINATORIAL OPTIMIZATION AND END-TO-END DEEP RL

Neural Combinatorial Optimization (NCO) learns routing policies using Pointer Networks [7], [24], reinforcement learning on dynamic customer states [9], Attention Models [8], and Graph Neural Networks [25], [26]. Extending NCO to VRPTW [10], [11] gives fast inference on small instances, but with the problems noted in Section I: decoding deadlocks under hard time windows, training on scalarized costs instead of fleet size first ($N_{\mathrm{V}} \succ D_{\mathrm{T}}$), and degradation on larger instances or different topologies.

C. LEARNING-AUGMENTED METAHEURISTICS

The "Learning to Optimize" line of work [27] embeds neural models inside metaheuristic search loops for operator selection [**?**], [14], solution perturbation [29], and learned destroy policies [15]. Closest to this work, Johnn et al. [**?**] formulate operator choice in ALNS as a Markov decision process

TABLE 1. Taxonomy of Representative Methodologies for VRPTW and Related Routing Problems

Methodology Category	Representative Works	Hard TW Feasible	Objective Paradigm	State-Aware Selection	Hierarchical Control	Learned Accept	Per-Step Compute Cost
Classical Metaheuristics	ALNS [5], [6]	✓	Lexicographic	× (Stateless)	×	× (SA)	Minimal (Heuristic)
	SISR [12], HGS [13]	✓	Lexicographic	× (Rule-based)	×	× (SA/Population)	Minimal (Heuristic)
End-to-End Neural Solvers	AM [8], DRL-VRP [9]	× (Penalized)	Scalarized	N/A	×	N/A	Very Low (One-shot)
	Neural VRPTW [10], [11]	× (Mask-fragile)	Scalarized	N/A	×	N/A	Very Low (One-shot)
Single-Policy Hybrid RL-LNS	L2I [14]	N/A (No TW)	Distance	✓	× (Single-agent)	× (None)	Moderate (Policy NN)
	Neural LNS [15]	N/A (No TW)	Distance	✓	× (Single-agent)	× (SA/Greedy)	Moderate (Policy NN)
	Graph RL ALNS [**?**], DR-ALNS [**?**]	Problem-dependent	Problem-dependent	✓	× (Single policy)	Partly (DR-ALNS)	Moderate (Policy NN)
Proposed Tri-Level Hybrid DDQN-ALNS	**This Work**	**Validated (Returned plan)**	**Lexicographic**	✓	✓ **(Two controllers)**	✓ **(LAC)**	**High (RL/MIP)**

solved with deep reinforcement learning and graph neural networks, and DR-ALNS [**?**] learns a policy that selects operators, adjusts parameters and controls the acceptance criterion, so learned influence on acceptance is not new. Our design differs in three respects: separate controllers for plateau regimes and operator pairs in place of one policy, an acceptance model added to Simulated Annealing by a disjunctive rule, and online learning from random weights in every run with no prior policy training for these controllers. We did not run these methods, and these differences are not evidence that our method performs better.

D. POSITIONING OF THIS WORK

Table 1 places Tri-Level Hybrid DDQN-ALNS in the routing literature. Our approach combines validated feasibility of returned solutions with two controllers at different timescales and a Learned Acceptance Criterion. Separating macro and micro control addresses credit assignment across timescales [41]: the reward for leaving a plateau of several hundred iterations comes long after any single destroy-repair decision. That motivates the separation, although we do not compare against a single flat policy. The macro tier's semi-MDP formulation over $\Delta_{\text{seg}} = 100$-iteration options separates plateau control from operator selection. Table 1 also shows the cost: online network updates and MIP solves consume part of each time budget. Section V measures whether this pays off.

III. PROBLEM FORMULATION

The Vehicle Routing Problem with Time Windows (VRPTW) is formally defined on a complete directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{A})$. The vertex set $\mathcal{V} = \mathcal{N} \cup \{0, n+1\}$ comprises customer nodes $\mathcal{N} = \{1, 2, \ldots, n\}$, origin depot node 0, and destination depot node $n+1$. The physical depot is modeled as two separate nodes (0 and $n+1$) to represent route departure and return without self-loops. The arc set is $\mathcal{A} = \{(i,j) : i, j \in \mathcal{V}, i \neq j, i \neq n+1, j \neq 0\}$.

Each customer $i \in \mathcal{N}$ has a positive demand $q_i > 0$, service duration $s_i \geq 0$, and service time window $[e_i, l_i]$, where e_i and l_i denote the earliest and latest allowable service start times, respectively. Depot nodes satisfy $q_0 = q_{n+1} = 0$, $s_0 = s_{n+1} = 0$, and $[e_0, l_{n+1}]$ defines the global planning horizon. Each arc $(i,j) \in \mathcal{A}$ has travel distance $c_{ij} \geq 0$ and travel time $t_{ij} \geq 0$. A homogeneous fleet of K vehicles, denoted by $\mathcal{K} = \{1, \ldots, K\}$, is stationed at the depot, with each vehicle possessing carrying capacity Q.

A. MATHEMATICAL MODEL AND DECISION VARIABLES

The optimization model employs binary routing variables $x_{ij}^k \in \{0,1\}$ (1 if vehicle $k \in \mathcal{K}$ traverses arc $(i,j) \in \mathcal{A}$, and 0 otherwise) and continuous schedule variables $w_i^k \geq 0$ specifying the time at which vehicle $k \in \mathcal{K}$ begins service at node $i \in \mathcal{V}$. The primary objective is lexicographic: first minimizing the number of deployed vehicles (N_{V}), and secondarily minimizing total travel distance (D_{T}):

$$\text{lex min} \left(\sum_{k \in \mathcal{K}} \sum_{j \in \mathcal{N}} x_{0j}^k, \quad \sum_{k \in \mathcal{K}} \sum_{(i,j) \in \mathcal{A}} c_{ij}\, x_{ij}^k \right) \tag{2}$$

In computational solver implementations, this hierarchy is evaluated via a scalarized surrogate objective using a large penalty $\mathcal{M} \gg \sum_{(i,j) \in \mathcal{A}} c_{ij}$:

$$\min \sum_{k \in \mathcal{K}} \left(\mathcal{M} \sum_{j \in \mathcal{N}} x_{0j}^k + \sum_{(i,j) \in \mathcal{A}} c_{ij}\, x_{ij}^k \right) \tag{3}$$

subject to:

$$\sum_{k\in\mathcal{K}}\sum_{j\in\mathcal{V},j\neq i} x_{ij}^k = 1, \quad \forall i \in \mathcal{N} \tag{4}$$

$$\sum_{j\in\mathcal{N}\cup\{n+1\}} x_{0j}^k \leq 1, \quad \forall k \in \mathcal{K} \tag{5}$$

$$\sum_{j\in\mathcal{N}\cup\{0\}} x_{j,n+1}^k = \sum_{j\in\mathcal{N}\cup\{n+1\}} x_{0j}^k, \quad \forall k \in \mathcal{K} \tag{6}$$

$$\sum_{i\in\mathcal{V},i\neq h} x_{ih}^k - \sum_{j\in\mathcal{V},j\neq h} x_{hj}^k = 0, \quad \forall h \in \mathcal{N},\ k \in \mathcal{K} \tag{7}$$

$$\sum_{i\in\mathcal{N}} q_i \sum_{j\in\mathcal{V},j\neq i} x_{ij}^k \leq Q, \quad \forall k \in \mathcal{K} \tag{8}$$

$$w_i^k + s_i + t_{ij} - M_{ij}(1 - x_{ij}^k) \leq w_j^k, \quad \forall (i,j) \in \mathcal{A},\ k \in \mathcal{K} \tag{9}$$

$$e_i \leq w_i^k \leq l_i, \quad w_i^k \geq 0, \quad \forall i \in \mathcal{V},\ k \in \mathcal{K} \tag{10}$$

$$x_{ij}^k \in \{0,1\}, \quad \forall (i,j) \in \mathcal{A},\ k \in \mathcal{K} \tag{11}$$

Constraint (4) ensures that every customer is visited exactly once. Constraints (5)–(6) enforce that each vehicle departs from origin depot 0 and arrives at destination depot $n+1$ at most once. Flow conservation across all customer nodes is guaranteed by (7). Constraint (8) prevents vehicle capacity violations. Schedule propagation is maintained via (9), where $M_{ij} = \max\{0, l_i + s_i + t_{ij} - e_j\}$ defines the tightest valid Big-M upper bound. Customer time-window compliance and domains are governed by (10)–(11); if a vehicle arrives at customer i prior to e_i, it incurs an idle waiting time such that service begins at $w_i^k = e_i$. Although w_i^k is indexed per vehicle to match x_{ij}^k, constraint (4) ensures that exactly one vehicle serves each customer i, so only one w_i^k is active in any feasible plan.

IV. PROPOSED METHODOLOGY

This section describes Tri-Level Hybrid DDQN-ALNS, which combines online reinforcement learning controllers with classical search for the lexicographic VRPTW objective.

A. TRI-LEVEL ARCHITECTURE OVERVIEW

The method has three tiers, shown in Fig. 1:

1) Macro Decision Tier (Plateau Controller π_{macro}): formulated as an aggregated semi-MDP over segments of length $\Delta_{\text{seg}} = 100$, where a Dueling Double DQN [38], [39] with Prioritized Experience Replay [40] observes search progress and switches among 7 search modes;
2) Micro Selection Tier (Operator Controller π_{micro} & LAC): operates per iteration t, using a Micro Dueling DDQN with a normalized Softmax-Entropy gate to select a destroy–repair operator pair from a 20-dimensional state, together with a Learned Acceptance Criterion (LAC) trained on delayed outcomes; and
3) Heuristic Search Tier (Granular ALNS & Route-Pool Recombination): executes ruin-and-recreate operations with $O(1)$-per-query Savelsbergh forward-slack insertion checks on cached route profiles, restricts moves to the $k = 15$ nearest customers, and periodically solves a time-limited Set Partitioning formulation over elite routes via HiGHS.

B. MACRO LEVEL: PLATEAU CONTROLLER (AGGREGATED SMDP FORMULATION)

We formulate the macro tier as a Semi-Markov Decision Process [41] over an Aggregated State Representation (Aggregated SMDP) defined by the tuple $(\mathcal{S}_{\text{macro}}, \mathcal{A}_{\text{macro}}, P_{\text{macro}}, R_{\text{macro}}, \gamma_{\text{macro}})$. Because the underlying combinatorial route pool and fine-grained trajectory history are projected into a 12-dimensional feature space $\mathbf{s}_k^{\text{macro}}$, the decision tier operates over a compact aggregated state representation with temporal options of fixed duration $\Delta_{\text{seg}} = 100$ low-level ALNS iterations. The controller acts only once the search has stagnated for a minimum number of iterations; before that, segments run in the default mode and are not used as macro transitions. The temporally extended transition kernel $P_{\text{macro}}(\mathbf{s}_{k+1}^{\text{macro}} \mid \mathbf{s}_k^{\text{macro}}, a_k)$ denotes the distribution of the next observation after the 100 underlying stochastic destroy–repair steps. The learner discounts per macro decision with $\gamma_{\text{macro}} = 0.95$, used both in the DDQN target and in the shaping term below, rather than with $\gamma^{\Delta_{\text{seg}}}$.

The aggregated formulation is a modeling device, not a verified Markov model. The 12 features are a lossy summary of the full search state, which also includes the current routes, the route pool, the operator weights, the network parameters and the random-number state. Distinct search states therefore share an observation, and the next observation depends on components that the controller does not see, so $\mathbf{s}_k^{\text{macro}}$ is not in general a sufficient statistic and the macro process need not satisfy the Markov property. The controller is better described as a learned policy over a partially observed, state-aggregated process [**?**]. We provide no convergence guarantee and no bound on the approximation error. Such guarantees are not available in general for Q-learning over aggregated or partial observations, and the online DDQN additionally uses nonlinear function approximation, experience replay and target networks. The value of the controller is therefore assessed empirically through the ablation study (Section V-F).

1) Macro State Representation

At the start of segment k, the macro controller observes a 12-dimensional state vector $\mathbf{s}_k^{\text{macro}} \in \mathbb{R}^{12}$, covering search progress, fleet size and route structure:

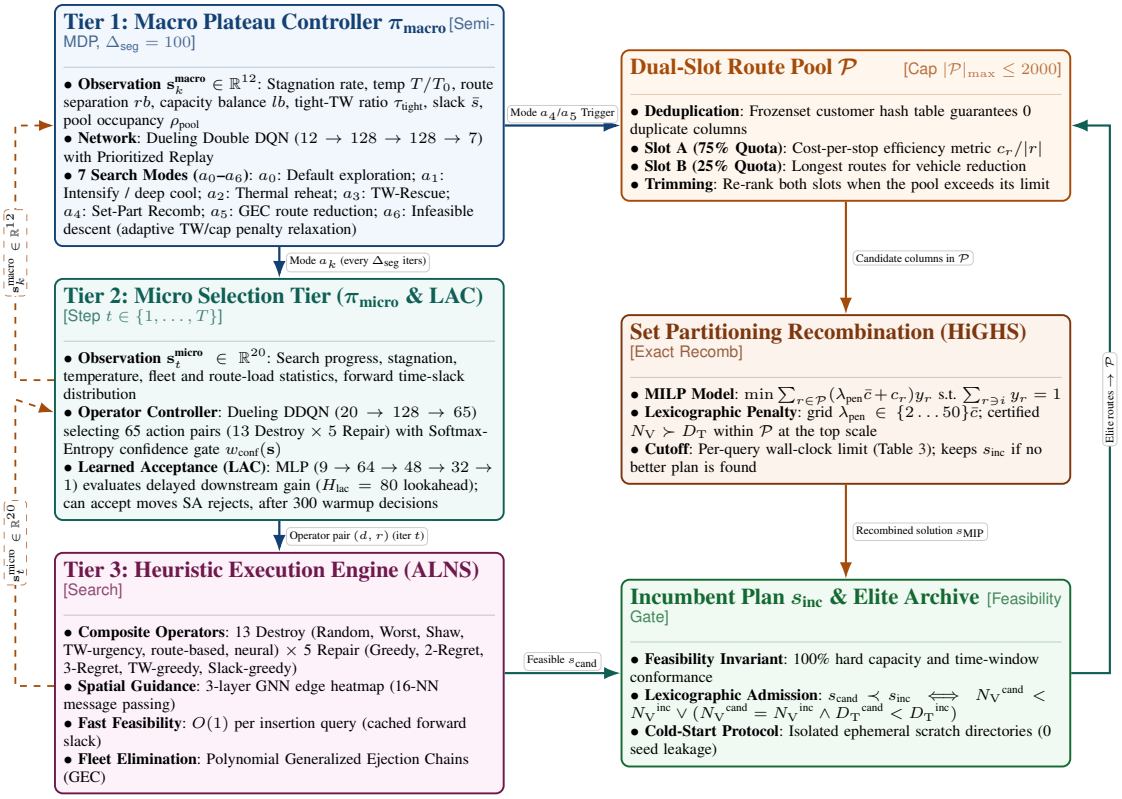


FIGURE 1. Tri-Level Hybrid DDQN-ALNS control architecture. Solid downward arrows represent hierarchical control and data dispatch across tiers. Dashed upward arrows represent state feedback signals consumed by the Macro and Micro RL training loops. The Macro tier operates over segment options $\Delta_{\text{seg}} = 100$; the Micro and Heuristic tiers execute per iteration t.

$$\mathbf{s}_k^{\text{macro}} = \Big[\min\Big(\frac{\text{no_imp}}{\text{patience}}, 1\Big),\ \min\Big(\frac{C_{\text{cur}} - C_{\text{best}}}{C_{\text{best}}}, 1\Big),\ \min\Big(\frac{T}{T_0}, 1.5\Big),\ \text{imp_rate},\ \min\Big(\frac{N_{\text{V,cur}}}{N_{\text{V,init}}}, 2\Big),\ rb,\ lb,\ \tau_{\text{tight}},\ \bar{s},\ \rho_{\text{fill}},\ \rho_{\text{pool}},\ \frac{k}{K_{\max}} \Big] \tag{12}$$

where no_imp is the number of consecutive unimproved iterations, T/T_0 is the normalized annealing temperature, $rb \in [0,1]$ and $lb \in [0,1]$ measure spatial route separation and capacity balance, $\tau_{\text{tight}} = \frac{1}{n}\sum_{i=1}^{n} \mathbb{I}(l_i - e_i < 0.5 \cdot d_{\max})$ is the fraction of customers whose time-window width is below half the largest distance $d_{\max}$ of the instance, $\bar{s}$ is the average forward time slack, ρ_{fill} is the average vehicle capacity utilization, and $\rho_{\text{pool}} = \min(|\mathcal{P}|/\mathcal{P}_{\max}, 1.0)$ tracks the route pool fill ratio.

2) Macro Action Space & Search Regimes

The macro action space $\mathcal{A}_{\text{macro}}$ comprises 7 search modes. Each mode scales the destroy size, multiplies the temperature once per segment, biases the operator priors and sets the local-search passes. With (destroy scale, temperature factor) in parentheses, the modes are: a_0 (DEFAULT, 1.00, 1.00); a_1 (INTENSIFY, 0.70, 0.98); a_2 (DIVERSIFY, 1.35, 1.08), which also permits a one-vehicle increase in the current solution; a_3 (TW_RESCUE, 1.10, 1.05), with priors favoring time-window urgency removal and time-window-aware repair; a_4 (POOL_RECOMBINE, 0.90, 1.01), which may trigger Set Partitioning; a_5 (ROUTE_REDUCE, 0.95, 1.02), with priors favoring route-elimination operators, a Generalized Ejection Chain attempt at every iteration, penalized acceptance, and possible recombination; and a_6 (INFEASIBLE_DESCENT, 1.10, 1.05), which is available only during the first search phase described below and maps to DEFAULT afterwards. The segment length is 100 iterations, or 130 on instances with wide time windows.

3) Search Phases

The search changes behavior over the budget. In the first phase, candidates are accepted by a penalized Simulated Annealing rule on $C_{\text{pen}}(s) = C(s) + \lambda\,(v_{\text{cap}}(s) + v_{\text{tw}}(s))$, so the current solution may be infeasible. The multiplier λ is updated every 100 iterations, by a factor of 0.90 when more than half of recent candidates are feasible and 1.15 otherwise, and is bounded by $[5, 200]$ for instances with tight time windows and $[0.5, 50]$ for others. The phase ends after 60% of a nominal 10,000 iterations, when the fleet reaches its floor, or, with no known floor, after 250 iterations without a new best solution (and at least 200 iterations in total). LAC is not consulted in this phase, and mode a_5 also keeps penalized acceptance afterwards.

The main destroy-repair loop receives 80% of the time

budget. The remaining 20% goes to tail phases: pool recombination, polishing at fixed fleet size, dense column generation for the pool, route-elimination heuristics, and distance polishing. ALNS-Base has no tail phases. A few settings also adapt to the instance: the initial temperature is scaled by 0.8 for tight and 1.2 for wide time windows, and the plateau trigger is 1.5 times longer for $N \geq 200$.

4) Multi-Objective Segment Reward & Surrogate Policy Optimization

The segment reward R_{seg} weighs vehicle reduction against distance savings through an adaptive fleet pressure factor $\lambda_{\text{fleet}} \in [0, 1]$:

$$R_{\text{seg}} = 0.30\big[\text{Base} + \lambda_{\text{fleet}}\Psi_{\text{NV}} + (1-\lambda_{\text{fleet}})\Psi_{\text{TD}} + \text{Bonus}_{\text{NV}}\big] + \Delta\Phi \tag{13}$$

where $\Psi_{\text{NV}} = \sum_{j=1}^{4} \beta_{\text{NV},j}\Delta X_j$ and $\Psi_{\text{TD}} = \sum_{j=1}^{4} \beta_{\text{TD},j}\Delta X_j$ linearly weight improvements across best-so-far and incumbent fleet size (ΔN_{V}) and travel distance (ΔC), with a representative primary fleet coefficient $\beta_{\text{NV},1} = 8.0$. The baseline term Base penalizes intensive local search passes and unproductive stagnation, while Bonus_{NV} provides a discrete reward of $+15$ per vehicle removed from the best solution and $+5$ per vehicle removed from the current solution. All coefficients are listed in Supplementary Section 6.

Equation (13) has two fleet rewards. The dense term ($\lambda_{\text{fleet}}\Psi_{\text{NV}}$) rewards moves that balance route loads and narrow the fleet gap to the incumbent. The sparse term (Bonus_{NV}) pays only when a vehicle is removed, so that signal is not diluted by routine changes in distance.

a: Reward Constants

The reward constants were set by hand during development, and Section **??** states what we can and cannot document about them and how the test set influenced later development.

Evaluation and training use different objectives. Incumbent replacement, elite archive admission and final selection use the lexicographic order $f(s) = (N_{\text{V}}(s), D_{\text{T}}(s))$, so fleet size always comes before distance. The policy instead receives the scalar reward R_{seg}, modulated by λ_{fleet}. When the current fleet is above the best ($N_{\text{V,cur}} > N_{\text{V,best}}$), $\lambda_{\text{fleet}} \to 1$ concentrates the reward on removing vehicles; once they match ($N_{\text{V,cur}} = N_{\text{V,best}}$), $\lambda_{\text{fleet}} \leq 0.5$ moves the weight to distance.

We also add potential-based reward shaping $\Delta\Phi = \gamma_{\text{macro}}\,\Phi(\mathbf{s}^{\text{macro}}_{k+1}) - \Phi(\mathbf{s}^{\text{macro}}_{k})$ to speed up learning without creating artificial reward cycles in the surrogate MDP. This shaping applies to the macro surrogate objective [30]. It does not give policy invariance with respect to the lexicographic VRPTW objective, which the evaluator enforces separately. The fleet pressure factor rises when the current fleet is above the best:

$$\lambda_{\text{fleet}}(s) = \sigma\left(8.0 \cdot \frac{N_{\text{V,cur}} - N_{\text{V,best}}}{N_{\text{V,init}}}\right) \tag{14}$$

where $\sigma(\cdot)$ is the logistic sigmoid, with the state potential defined as:

$$\begin{aligned}\Phi(s) = &-15\,\lambda_{\text{fleet}}(s)\frac{\max(N_{\text{V}} - N_{\text{V,best}}, 0)}{N_{\text{V,init}}} \\ &- 0.18\big(1 - \lambda_{\text{fleet}}(s)\big) \\ &\times \text{clip}\left(\frac{D_{\text{T}} - D_{\text{T,best}}}{D_{\text{T,best}}} \cdot 100, -25, 25\right)\end{aligned} \tag{15}$$

The Macro Plateau Controller trains its own Dueling Double DQN (Q_{macro}) with prioritized replay buffer $\mathcal{D}_{\text{macro}}$, taking Adam steps with a cosine-annealed learning rate at the end of each macro segment.

C. MICRO LEVEL: OPERATOR CONTROLLER & ACTION BLENDING

1) Micro State Representation

At each iteration t, the operator controller observes a 20-dimensional state vector $\mathbf{s}^{\text{micro}}_t \in \mathbb{R}^{20}$ (Table 2), which describes search progress and route structure.

2) Composite Action Space

The micro action space $\mathcal{A}_{\text{micro}}$ comprises $N_D \times N_R = 13 \times 5 = 65$ composite heuristic pairs, combining 13 Destroy Operators (Random, Worst-cost, Shaw spatiotemporal relatedness, Route portion removal, Time-window urgency removal, Route-Depletion Destroy, Route dispersion elimination, Costly route elimination, Cross-route Shaw, Route merge sampling, Route absorb disruption, Neural worst, Neural Shaw) with 5 Repair Operators (Greedy insertion, 2-Regret, 3-Regret, Time-window greedy, Forward-time-slack greedy).

3) Dueling DDQN Architecture & Policy-Concentration Blending

The Micro Q-network adopts a Dueling architecture [31] with layer normalization:

$$Q(s,a;\theta) = V(s;\theta_v) + \left(A(s,a;\theta_a) - \frac{1}{|\mathcal{A}|}\sum_{a'} A(s,a';\theta_a)\right) \tag{16}$$

A normalized Softmax-Entropy Policy-Concentration Gate balances the Q-values against the exploration priors. Let $\pi_\theta(a \mid s) = \text{softmax}(Q(s,a)/\tau_{\text{soft}})$ be the Boltzmann policy

TABLE 2. 20-Dimensional Micro State Vector $\mathbf{s}^{\text{micro}}_t$

Idx	Sym	Definition	Idx	Sym	Definition
s_1	ΔC_{rel}	$(C_{\text{cur}} - C_{\text{best}})/C_{\text{best}}$	s_{11}	ρ_{fill}	Capacity fill ratio
s_2	Ratio_{NV}	$N_{\text{V,cur}}/N_{\text{V,init}}$	s_{12}	ρ_{pool}	Pool fill ratio $\lvert\mathcal{P}\rvert/\mathcal{P}_{\max}$
s_3	$\text{Prog}_{\text{global}}$	$\min(t/10000, 1)$	s_{13}	m_{idx}	Active macro mode index
s_4	Prog_{seg}	$(t \bmod \Delta_{\text{seg}})/\Delta_{\text{seg}}$	s_{14}	$\Delta N_{\text{V,rel}}$	$(N_{\text{V,cur}} - N_{\text{V,best}})/N_{\text{V,init}}$
s_5	T_{norm}	T/T_0	s_{15}	Imp_{rec}	Recent improvement rate
s_6	$\text{Stag}_{\text{ratio}}$	no_imp/patience	s_{16}	Gini_{len}	Gini of route loads
s_7	rb	Route separation index	s_{17}	μ_{slack}	Mean customer forward slack
s_8	lb	Capacity load balance	s_{18}	σ_{slack}	Std of forward slacks
s_9	τ_{tight}	Frac tight TW customers	s_{19}	Frac_{cap}	Fraction routes $> 90\%$ cap
s_{10}	$\bar{s}$	Normalized mean slack	s_{20}	Var_{dist}	Normalized variance of route distances

at softmax temperature $\tau_{\text{soft}} = 1.0$. The normalized Shannon entropy is defined as:

$$\mathcal{H}(\pi_\theta(\cdot \mid s)) = -\frac{1}{\ln|\mathcal{A}|}\sum_{a\in\mathcal{A}} \pi_\theta(a \mid s)\ln \pi_\theta(a \mid s) \in [0,1] \tag{17}$$

We define the policy-concentration weight as $w_{\text{conf}}(s) = 1.0 - \mathcal{H}(\pi_\theta(\cdot \mid s))$. High concentration ($w_{\text{conf}} \to 1$) indicates that the Boltzmann policy is sharply peaked around a dominant operator pair, whereas low concentration ($w_{\text{conf}} \to 0$) signals an exploratory regime.

The action-selection score blends the Q-values with operator priors and the mean of a Thompson-sampling posterior through this gate, then adds a variance-aware bonus adapted from UCB [37] and UCB-V [36]:

$$\begin{aligned} Q_{\text{blend}}(s,a) &= w_{\text{conf}}(s)\, Q(s,a) + (1 - w_{\text{conf}}(s))\big[\alpha_{\text{p}} \ln P(a) \\ &\quad + \beta_{\text{b}}\, \bar{\mu}_{\text{bandit}}(a)\big], \\ Q_{\text{eff}}(s,a) &= Q_{\text{blend}}(s,a) + \alpha_{\text{u}}\, \frac{U(a) - \bar{U}}{\sigma_U}, \\ U(a) &= \hat{\mu}(a) + c_{\text{ucb}}\, \sigma_a \sqrt{\ln N_t / N_a(t)}, \end{aligned} \tag{18}$$

where $\alpha_{\text{p}} = 0.55$ scales prior log-probabilities, $\beta_{\text{b}} = 0.20$ weights the bandit posterior mean, $\alpha_{\text{u}} = 0.35$ weights the standardized bonus over the actions, $c_{\text{ucb}} = 1.0$, $\hat{\mu}(a)$ is a running mean reward, and $\sigma_a = \sqrt{M_{2,a}/\max(N_a(t) - 1, 1)}$ is an online standard deviation tracked with Welford's algorithm. Action counters start at 0.5 and decay by 0.993 per step, and $N_t = \sum_a N_a(t)$.

Two properties of this rule follow directly. The Q-values and log-priors are not standardized before the gate, so the balance between them depends on the scale of the learned values. The bonus is not gated, so it continues to affect selection when the policy is concentrated.

The first 256 transitions select pairs from the bandit with the prior. After that, the controller explores with probability ε, sampling from the prior and bandit, with ε decaying from 0.35 to 0.02 by a factor of 0.9996 per step. It otherwise takes the pair that maximizes Q_{eff}. The Micro reward has the same form as R_{seg} at iteration level, with its own coefficients and a scale of 0.45; a rejected candidate receives a base reward of -0.08 before shaping.

D. LEARNED ACCEPTANCE CRITERION (LAC): SA WITH PREDICTIVE LOOKAHEAD OVERRIDE

In classical ALNS, move acceptance relies on Simulated Annealing (SA), which accepts improving transitions ($\Delta C < 0$) deterministically and non-improving transitions ($\Delta C \geq 0$) with probability $P_{\text{SA}} = \exp(-\Delta C/T)$. During deep cooling stages ($T \to 0$), P_{SA} vanishes exponentially, restricting search to downhill moves, so it can get stuck in local optima.

To reduce this trapping, the Learned Acceptance Criterion (LAC) adds an online supervised binary classifier $\phi_{\text{LAC}} : \mathbb{R}^9 \to [0,1]$. LAC is consulted only for feasible candidates that keep the current fleet size and increase travel distance; improving, fleet-reducing and infeasible candidates follow the rules of Algorithm 1 unchanged, and penalty-relaxed phases use the penalized SA rule without LAC. For these candidates, LAC does not filter or replace SA. It adds a disjunctive rule:

$$\begin{aligned} \text{Accept}(s' \mid s, T) \iff &\Big(p_t \geq \tau_{\text{acc}}(t)\Big) \\ &\vee \Big(P_{\text{SA}}(\Delta C, T) > U\Big) \end{aligned} \tag{19}$$

where $p_t = \phi_{\text{LAC}}(\mathbf{z}_t)$, $U \sim \text{Uniform}(0,1)$, and $\tau_{\text{acc}}(t) \in [0.40, 0.75]$ is an adaptive confidence threshold.

The disjunction ($\vee$) means LAC can only add acceptances. Any consulted move that standard Simulated Annealing would accept ($P_{\text{SA}} > U$) is also accepted by rule (19). When SA would reject a non-improving move ($P_{\text{SA}} \leq U$), the move is still accepted if the classifier predicts a later incumbent improvement with probability $p_t \geq \tau_{\text{acc}}(t)$. These extra transitions mean that rule (19) defines a different Markov transition kernel from SA, and the classical asymptotic convergence results for SA under logarithmic cooling [**?**] do not carry over. We make no convergence claim for the modified kernel: the search uses geometric cooling within a finite budget, and the contribution of LAC is assessed empirically (Section V-F). The supervision target is observational. The first $W_{\text{lac}} = 300$ consulted decisions use pure SA while the classifier collects training data; for every consulted move, including warmup moves, the label records whether the global best travel distance improved within the next $H_{\text{lac}} = 80$ LAC decisions (not 80 ALNS iterations), irrespective of whether the move was accepted. The label therefore indicates a promising search context rather than the causal effect of accepting that particular move.

Because rule (19) only adds acceptances, it increases the number of accepted non-improving moves during deep cooling. Sections V-F and VI-D return to the difficulty of separating useful lookahead from a higher acceptance rate.

1) Features and Training

The LAC input feature vector $\mathbf{z}_t \in \mathbb{R}^9$ comprises 9 normalized features:

$$\begin{aligned} \mathbf{z}_t = \Big[&\frac{\Delta C}{|C_{\text{cur}}|},\ \frac{T}{T_0},\ \frac{\text{no_imp}}{\text{patience}}, \\ &\text{clip}(\Delta N_{\text{V}}, -2, 2),\ \min\Big(\frac{t}{10000}, 1\Big), \\ &\tau_{\text{tight}},\ \rho_{\text{fill}},\ \bar{s},\ \exp\left(-\frac{\max(\Delta C, 0)}{T}\right)\Big] \end{aligned} \tag{20}$$

During the first 300 consulted decisions, the solver uses Simulated Annealing alone while queuing the features of each decision. After a further $H_{\text{lac}} = 80$ consulted decisions, each queued decision receives a binary label: $y_t = 1$ if the best distance C_{best} has decreased since that decision, and 0 otherwise. The label therefore follows distance, not the lexicographic objective, so a fleet reduction that raises the best distance receives $y_t = 0$. The threshold rises with progress $p = \min(t/10000, 1)$ as $\tau_{\text{acc}}(t) = 0.40 + 0.35\, p^{0.75}$.

Because training examples are labeled online from the active search trajectory, the empirical training distribution is endogenous and non-stationary. Every 20 consulted decisions, LAC trains a 4-layer MLP ($9 \to 64 \to 48 \to 32 \to 1$, with LayerNorm and 0.1 dropout after the first layer) on a batch of 64 from a buffer of 5,000 labeled decisions, using Adam (lr $= 10^{-3}$) with class-weighted binary cross-entropy loss:

$$\mathcal{L}_{\text{LAC}}(\phi) = -\frac{1}{B}\sum_{i=1}^{B}\Big[w_1 y_i \log p_i + (1-y_i)\log(1-p_i)\Big] \tag{21}$$

where $p_i = \phi_{\text{LAC}}(\mathbf{z}_i)$ and $w_1 = N_{\text{neg}}/N_{\text{pos}}$ balances positive and negative labels within the batch.

E. HEURISTIC SEARCH COMPONENTS

The low-level search combines cached insertion checks, granular candidate lists, a frozen GNN heatmap and ejection chains.

1) Constant-Time Insertion Feasibility Queries via Savelsbergh Forward Slack

To avoid an $O(m)$ recomputation of route timing for each candidate insertion, we maintain Savelsbergh forward time slack F_i for each node i along route $R = (v_0, v_1, \ldots, v_m)$ [33]:

$$F_i = \min_{i \le j \le m}\left\{ l_j - w_j + \sum_{k=i+1}^{j} W_k \right\} \tag{22}$$

A candidate insertion of customer u between positions i and $i+1$ inducing arrival delay Δt_u at node $i+1$ preserves temporal feasibility for all subsequent customers if and only if $\max(0, \Delta t_u - W_{i+1}) \le F_{i+1}$, where W_k is the waiting time at node k, given that the prefix up to v_i is feasible and u is served within its window; capacity feasibility additionally requires $q(R) + q_u \le Q$. Given a cached timing profile of the route (arrival times, load, and forward slacks or, equivalently, the latest feasible service start of each node, which our implementation stores), each such query costs $O(1)$. The arrival time delay shifted to customer v_{i+1} is computed as:

$$\Delta t_u = \max\big(0, e_u - (w_i + s_i + t_{i,u})\big) + t_{i,u} + s_u + t_{u,i+1} - t_{i,i+1} \tag{23}$$

where w_i is the service start time at node i, s denotes service durations, and e_u is the earliest ready time. The $O(1)$ bound applies to a single insertion query and holds only while the cached profile is valid. Building or refreshing the profile costs $O(m)$ for a route of length m and is required after every accepted modification of that route; scanning all $m+1$ insertion positions of one customer is therefore $O(m)$ per route, and evaluating one customer against K routes is $O(Km)$ before granular pruning. Moves that reverse or relocate a segment (2-opt, ejection chains) recompute arrival times along the modified segment, so their cost grows with the segment length, and only the suffix check after the segment uses the cached slack. We claim no $O(1)$ bound for complete moves or iterations.

2) Granular Correlated Neighborhood Search (GNS)

Following granular search principles [34], inter-route operators restrict candidate evaluations to the $k = 15$ nearest customers of each customer by travel distance, precomputed once per instance with the depot excluded. This bounds the candidate insertion search space to $O(kN)$ move queries per iteration.

3) Lightweight Graph Neural Network for Spatial Edge Guidance

A lightweight Graph Neural Network (GNN) adds a learned edge probability heatmap $\mathbf{H} \in [0,1]^{(N+1)\times(N+1)}$ to the heuristic candidate lists. The graph encodes $N+1$ nodes (depot + customers) with 6-dimensional normalized node features $\mathbf{v}_i \in \mathbb{R}^6$:

$$\mathbf{v}_i = \left[\tilde{x}_i,\ \tilde{y}_i,\ \frac{q_i}{Q},\ \frac{e_i}{T_{\text{depot}}},\ \frac{l_i}{T_{\text{depot}}},\ \frac{s_i}{T_{\text{depot}}}\right] \tag{24}$$

where $(\tilde{x}_i, \tilde{y}_i) \in [0,1]^2$ are min-max normalized Cartesian coordinates, q_i/Q is demand normalized by vehicle capacity, and (e_i, l_i, s_i) are ready time, due time, and service duration normalized by the depot operational horizon T_{depot}. Message passing runs over sparse k-nearest-neighbor edge sets ($K = 16$, with slot 0 pinned to the depot) with 1-dimensional edge feature $e_{ij} = c_{ij}/c_{\max}$ representing normalized Euclidean distance.

The network architecture comprises a linear embedding layer ($d_h = 64$), 3 sparse joint node-edge message-passing layers with LayerNorm and residual connections, and a bilinear projection head:

$$\mathbf{H}_{ij} = \sigma\left(\mathbf{z}_i^{\top}\mathbf{W}_{\text{edge}}\mathbf{z}_j\right) \tag{25}$$

producing edge selection probabilities $\mathbf{H}_{ij} \in [0,1]$. Message passing uses the 16 nearest neighbors of each node. The registered checkpoint was trained offline with a weighted binary cross-entropy loss and sampled negative edges on 48 synthetic instances (16 each of the C, R and RC distributions, half with $N = 100$ and half with $N = 200$), with 12 further synthetic validation instances of the same sizes. Training targets are the route edges of a 10-second ALNS run without the GNN, so they are heuristic labels and not optimal edges. The checkpoint was selected by minimum validation loss (epoch 71 of 150, Adam, learning rate 10^{-3}, one training seed), and no benchmark test instance was used for training or selection. On the 12 validation instances, thresholds of 0.05, 0.01 and 0.003 retain 23.3%, 29.6% and 34.1% of candidate arcs and 99.45%, 99.90% and 99.90% of teacher edges. These instances also selected the checkpoint, so the figures are optimistic, and they measure agreement with heuristic labels, not a routing gain. The effect of the GNN on solutions is estimated only by the factorial ablation. $N = 400$ lies outside the training sizes, so use at that size is an extrapolation. During search, network weights are frozen, and inference is executed once per instance during preprocessing in $O(NK)$ time, populating a static heatmap $\mathbf{H}$ that guides repair and filters insertion positions.

4) Generalized Ejection Chains (GEC)

To attack vehicle-count plateaus, a 3-level ejection chain tries to eliminate the shortest route. It is attempted at every iteration in mode a_5, every fourth iteration when a route has at most four customers, and every twentieth consecutive iteration without improvement. Customers of the target route are processed by decreasing demand, and each displaced customer u triggers a search over all other routes:

(i) Level 1 (Direct Insertion): insert u into another route R_j at minimum added distance;
(ii) Level 2 (Single-Customer Displacement): eject $v \in R_j$, insert u into R_j, and insert v into a third route R_k; and
(iii) Level 3 (Two-Customer Chained Displacement): eject $v \in R_j$ and insert u; eject $w \in R_k$ and insert v; and insert w into a further route. This level runs only when at most 30 routes remain.

The search has no candidate beam and no internal time check, so the solver can stop it only after it returns, while the supervisor enforces the cutoff independently. Each feasibility test replays the modified route at linear cost, not the $O(1)$ cached query above. For m displaced customers, K routes and maximum route length L, the worst-case work is $O(mK^3L^4)$ (Supplementary Section 3).

5) Feasibility Invariant Across Operational Layers

Feasibility differs by operational level: (1) Current solution (s_{cur}): may be infeasible under penalized acceptance (Section **??**); (2) Candidate moves (s_{cand}): violations are penalized in those phases and rejected otherwise; (3) Best solution (s_{inc}): only solutions with full capacity and time-window compliance can update it; (4) Elite archive and (5) Route pool: hold only feasible solutions and routes; and (6) Returned solution (s^*): counted only after the supervisor validates it in the original problem space (Section **??**).

F. MATHEURISTIC RECOMBINATION: TIME-LIMITED HIGHS SET PARTITIONING

During search stagnation, elite feasible routes accumulated in Route Pool $\mathcal{P}$ are recombined by solving a time-limited Set Partitioning integer linear programming formulation:

$$\min \quad \sum_{r\in\mathcal{P}} (\lambda_{\text{pen}}\bar{c} + c_r)\, y_r \tag{26}$$

$$\text{s.t.} \quad \sum_{r\in\mathcal{P}:i\in r} y_r = 1, \quad \forall i \in \mathcal{N} \tag{27}$$

$$y_r \in \{0,1\}, \quad \forall r \in \mathcal{P} \tag{28}$$

where c_r is route travel distance, $\bar{c}$ is pool mean route distance, and $\lambda_{\text{pen}} \in \{2.0, 5.0, 12.0, 30.0, 50.0\}$ is a penalty multiplier; the vehicle penalty $\lambda_{\text{pen}}\bar{c}$ has a floor of 200 in the implementation. At most 800 columns of the pool enter one model, and a vehicle ceiling constraint may be added. If exact partitioning fails, the solver falls back to a set-covering relaxation with duplicate customers removed afterwards.

Proposition 1 (Penalty Bound for Lexicographic Fleet Preservation): *Let $\mathcal{Y}_{SP} \subseteq \{0,1\}^{|\mathcal{P}|}$ be the set of feasible partitions over route pool $\mathcal{P}$. If $\lambda_{pen}\bar{c} > \sup_{\mathbf{y}_1,\mathbf{y}_2\in\mathcal{Y}_{SP}} \left(\sum_{r\in\mathbf{y}_1} c_r - \sum_{r\in\mathbf{y}_2} c_r\right)$, any partition $\mathbf{y}_1$ with $\sum_r y_{1,r} < \sum_r y_{2,r}$ strictly achieves a lower objective value than $\mathbf{y}_2$, guaranteeing exact lexicographic fleet minimization priority over distance within the column subset $\mathcal{P}$.*

With Route Pool $\mathcal{P}$ limited to $|\mathcal{P}| \le 2000$ columns, the distance difference between two valid partitions is bounded by $K_{\max} \cdot c^{\text{route}}_{\max}$. To satisfy Proposition 1 at every size, including $N = 400$, the solver applies the certified threshold $\lambda^{\text{cert}}_{\text{pen}} = \max\left(\lambda_{\text{pen}}\bar{c},\ (1+\epsilon_{\text{margin}})K_{\max}c^{\text{route}}_{\max}\right)$ with clearance margin $\epsilon_{\text{margin}} = 0.20$ at the largest scale of the multi-scale penalty grid $\lambda_{\text{pen}} \in \{2.0, 5.0, 12.0, 30.0, 50.0\}\bar{c}$ (formal derivation in Supplementary Material Proposition 1). The smaller scales are tried first and do not, in general, satisfy the condition of Proposition 1, so the partition they return need not be lexicographically optimal within $\mathcal{P}$; every partition is, however, subject to the hard fleet ceiling of the query. Proposition 1 therefore concerns lexicographic ordering among partitions of the route pool at the certified scale only. It does not imply optimality of the partition for the original VRPTW, whose optimal routes need not be in $\mathcal{P}$.

Across the whole search, the hierarchical objective is enforced at the level of the incumbent rather than at every transition. The current solution may temporarily accept a one-vehicle increase in the diversification mode, and penalty-relaxed phases may visit infeasible states. However, a candidate replaces the best solution only if it has fewer vehicles, or the same number of vehicles and strictly lower distance, after independent validation of exact cover, capacity and time windows in the original instance. Consequently the reported solution is non-increasing in N_{V} over time and never trades a vehicle for distance, but no claim is made that every intermediate move respects the ordering.

1) Route Pool Deduplication and Capacity Management

Routes visiting identical customer sets $\text{cover}(R) = \{v_1, \ldots, v_m\}$ are deduplicated with hash tables, which keep only the shortest tour. Pool capacity $|\mathcal{P}|_{\max} = \min(2000, 600 + 4N)$ is partitioned into Slot A (75% capacity, sorting by cost-per-stop $c_r/|r|$) and Slot B (25% capacity, preserving longest sub-routes $|r|$ to supply high-occupancy columns for fleet reduction), with at most $M_{\text{cust}} = \min(60, 28 + \lfloor N/10 \rfloor)$ routes per customer. HiGHS [32] solves the models by branch and cut. Each penalty scale is one query, limited to $\max(1, \min(30, 4 + 0.05N)/3)$ s (3, 4.7 and 8 s at $N = 100$, 200 and 400) and capped by the remaining budget.

G. GUIDED EJECTION SEARCH PHASE

An optional phase runs the route-minimization heuristic of Nagata and Bräysy [18] once, before the main search, for 10% of the remaining budget. It removes a random route and places its customers in an ejection pool. It then takes customers from the pool one at a time and inserts each one

Algorithm 1 Tri-Level Hybrid DDQN-ALNS Overview

Require: Instance $\mathcal{I}$, Hyperparameters Θ, Time budget T
Ensure: Best feasible routing plan s^*
$s^* \leftarrow s_{\text{inc}} \leftarrow \text{GreedyInit}(\mathcal{I})$; $\mathcal{P} \leftarrow \text{ExtractRoutes}(s^*)$
Initialize Macro DDQN Q_{macro}, Micro DDQN Q_{micro}, LAC ϕ_{LAC}
for $t = 1, 2, \ldots$ until 80% of T has elapsed **do**
  if $t \bmod \Delta_{\text{seg}} == 1$ **then** ▷ Macro Tier: SMDP regime selection
    Update regime $a_k \leftarrow \pi_{\text{macro}}(\mathbf{s}_k^{\text{macro}})$; trigger GEC/HiGHS if selected
  end if
  Select $(d, r) \leftarrow \pi_{\text{micro}}(\mathbf{s}_t^{\text{micro}})$ via confidence gating (18)
  $s_{\text{cand}} \leftarrow \text{GranularLocalSearch}(\text{Repair}(\text{Destroy}(s_{\text{cur}}, d), r), k = 15)$
  Accept s_{cand} by the phase's rule: penalized SA, or SA or LAC (ϕ_{LAC})
  Update route pool $\mathcal{P}$, train DDQNs and LAC on mature experiences
end for
Run tail phases until T: recombination, route elimination, polishing
return s^* ▷ Certified by the supervisor

if a feasible insertion exists. Otherwise it increments the customer's penalty counter, applies the feasible ejection of at most $k_{\max} = 5$ customers from one route that has the smallest penalty sum, and perturbs the solution with 100 random feasible relocate, exchange and 2-opt* moves. A route is eliminated when the pool empties, and the attempt restarts after 20,000 steps otherwise. The phase ends at its time limit or at the capacity bound $\lceil \sum_i d_i / Q \rceil$, and it reads no reference fleet. If it reduces the fleet, the result receives one local-search pass before the main search. This is a classical method, so any gain from it is not a gain from learning.

V. EXPERIMENTAL EVALUATION

This section evaluates the method on 74 standard benchmark instances: Solomon-100 (56 instances across 6 families) [2], Gehring-Homberger 200-customer (12 instances) and Gehring-Homberger 400-customer (6 instances) [35]. Five questions structure the evaluation:

- RQ1 (Comparison at equal compute): At 300 s, how does the method compare with ALNS-Base, PyVRP-ILS and adapted NeuroLKH on fleet size and, at matched fleet size, on distance?
- RQ2 (Budget): How do these comparisons change at 10, 60 and 900 s?
- RQ3 (Components): Which components change results, and does the learning bundle add to the heuristic bundle?
- RQ4 (Scale): How do the comparisons differ at 100, 200 and 400 customers?
- RQ5 (Families): How do they differ across clustered, random and mixed instance families? This question is descriptive.

A. EXPERIMENTAL SETUP, BASELINE FAIRNESS, AND COLD-START PROTOCOL

All experiments ran on one Apple M1 notebook (4 performance and 4 efficiency cores, 16 GB memory, macOS 26.5.2), with four cells at a time and one numerical-library thread per cell. macOS assigns cores itself, so processes were not pinned to cores.

Machine-speed variation. A wall-clock budget does not fix the compute a run receives. PyVRP and NeuroLKH with the same seed retrace the same search, so the time needed to reach the same (fleet, distance) step measures machine speed. On 18 instances at 60 s, cells of our first full sweep took a median of 2.3 times as long as the same cells on a later night, ranging from about 0.65 to 9.6 times per instance. During that sweep the notebook was also used for other work and was at times unplugged or asleep, but the cause of the slowdown is not established. Absolute times therefore do not describe a steady machine speed, and comparisons are valid within a block of cells that ran together, not across blocks. In the reported rerun, all cases of one instance, seed and cutoff run together as a block. A cell whose wall-clock span exceeds its monotonic duration by more than 5 s was frozen by machine sleep, and its whole block runs again. Each cell records power state and load at its start, and the primary comparisons are repeated without blocks that started a cell at a one-minute load above 8. Other work on the notebook raised the load as high as 233 early in the rerun. Before any result was read, we therefore moved 58 of the first 111 blocks at 300 s, those with a cell started above load 8, aside unread and ran them again, and from then on no new cell started above that load. A cell can still start below that load and then run above it, so also before any result was read, we began sampling load once a minute and added two repeats at 300 s: one without blocks that have a cell whose mean load during its run was above 8, and one that also leaves out blocks with a cell not measured this way. Load during a run is not known for the roughly 450 cells at 300 s that finished before sampling began.

ALNS-Base is our own implementation in the framework of Ropke and Pisinger [5], [6], not a reproduction of it. It shares the 13 destroy and 5 repair operators of our search infrastructure. It loads no GNN, so its two heatmap-guided destroy operators receive no heatmap. Pairs are selected by Thompson sampling over 65 Beta posteriors instead of a roulette wheel, with scores $(\sigma_1, \sigma_2, \sigma_3) = (33, 9, 3)$ normalized by σ_1, posterior decay 0.95 every 100 iterations, and a posterior reset after 1,250 iterations without improvement. Simulated Annealing starts at $T_0 = 0.05\, c_0 / \ln 2$, where c_0 is the cost of the starting solution, and cools geometrically with $\alpha = 0.99975$. Its main loop applies no local search, ejection chains or set partitioning. Initialization is algorithm-specific: ALNS-Base starts from the unweighted greedy construction (BuildGreedyInitialSolution), the hybrid starts from the same construction with GNN-weighted insertion costs, and PyVRP-ILS and NeuroLKH use their own construction procedures; construction and model inference are charged to each solver's budget. A defect in the greedy construction, which did not check the depot return deadline, made the ALNS-Base start infeasible on R1_2_5, so all 20 ALNS-Base runs on that instance terminated without a solution. We corrected the construction, which leaves the start unchanged on the other 73 instances, and reran those 20 runs.

PyVRP-ILS is the iterated local search of PyVRP [**?**], version 0.14.1, with a vehicle fixed cost set above any possible

travel cost so that its objective follows the fleet-first order. PyVRP's HGS is not the method used here. NeuroLKH-adapted runs the pretrained CVRPTW checkpoint and LKH binary of NeuroLKH [**?**] at a pinned upstream commit, with cold restarts at decreasing fleet sizes. It is an adaptation, not a replication of the authors' experiments, and we did not retrain it or tune it on benchmark instances. Both baselines work on integer-scaled models, so every candidate is mapped back and certified in the original problem space.

All cases share one supervisor. A case's clock starts in a fresh process with an empty work directory and JIT cache, and construction, model inference, search, JIT compilation after the start, certification and communication with the supervisor are charged to its budget. Imports and loading of frozen models are recorded separately. An incumbent counts only if the supervisor has received, validated and hashed it before the cutoff, and the process must return within $[T - 0.05, T + 0.5]$ s of the cutoff T.

a: Reference-Fleet Information

In the first full sweep (Section **??**), the hybrid read the best-known fleet size $N_{\mathrm{V,BKS}}$ [42] of each benchmark instance during search. It served as the floor at which vehicle-reduction phases stop, as the trigger for switching set partitioning to a distance-only objective, and as a parameter of the recombination and route-elimination steps. None of the baselines uses this information, it is not available for new instances, and equal compute alone does not remove this asymmetry. The rerun therefore disables every lookup. With the Guided Ejection Search phase on, the capacity bound takes the place of the reference fleet; with it off, the hybrid uses its fallbacks for unknown instances. The effect of reading the reference fleet was not re-estimated under matched conditions, so we report only that the first sweep's hybrid read it.

b: Hyperparameter Specification and Experimental Firewall

The hyperparameters in Table 3 combine literature defaults (for example the Adam optimizer parameters and the Polyak update rate $\tau = 0.005$) with values set by hand during development. We did not retain a calibration log, so we cannot document how the hand-set values were chosen or show that they are independent of earlier runs on the benchmark instances. They were frozen before the sweeps reported here. One solver rule reads family information from the benchmark's naming convention: the fleet-reduction search receives more iterations when an instance name begins with RC and the current fleet exceeds the target by at least one vehicle.

The test set did not, however, remain untouched by development. After the first full sweep, per-family losses on the test instances led us to evaluate a Guided Ejection Search phase before the main search. This is a classical method, so any gain from it is not a gain from learning. Its parameters were chosen only on 30 development instances with different instance identities: 18 are 100-customer subsamples of Gehring-Homberger instances, and 12 are unmodified Gehring-Homberger instances outside the test set. They are not independent of the test set, because they share customer coordinates with test instances of the same families and up to 81% of a development instance's customer records (coordinates, demand, time window and service time) are identical to those of a test instance. The rule for adopting the phase was fixed before a single contemporaneous rerun on the test set, and that rerun was not used to change the solver.

Training and testing protocol. The Macro DDQN, Micro DDQN and LAC have no offline training phase. They start from random weights in every run and learn online within that run's time budget, so they have no train/test split, nothing learned in one run is carried to another, and we make no claim that a learned policy transfers between instances. Only the GNN is trained offline. It is then frozen and uses only the synthetic data described in Section IV-E3. Its generalization is limited to what the factorial ablation measures on the benchmark instances.

The protocol has four rules for keeping runs independent and reproducible:

1) Cold starts: each solver run uses its own temporary directory (ColdStartScope), deleted when the run ends, and the Macro DDQN, Micro DDQN and LAC start from random weights;
2) Deterministic RNG Seeding: each worker process seeds Python's random, numpy, and PyTorch (torch.manual_seed and torch.cuda.manual_seed_all), using 5 independent seeds ($\{42, 43, 44, 45, 46\}$);
3) Objective Hierarchy: fleet size is the primary endpoint, and distance is compared only between seed pairs with matched fleet size (Section **??**); and
4) Recorded configuration: every run writes JSON metadata with resolved_components, which lists the modules that were active.

B. STATISTICAL ANALYSIS

Estimands and intervals follow a design fixed before the sweeps. The unit is a seed pair within an instance, not a best run or an instance mean. For each pair, the fleet score is $+1$ if the proposed configuration uses fewer vehicles, -1 if it uses more, and 0 if they tie; a case with a solution beats a case without one, and two cases without a solution tie. Scores are averaged over seeds and instances within each size and combined with fixed weights of $56/74$, $12/74$ and $6/74$. The score therefore lies in $[-1, 1]$ and is the net share of seed pairs won. It is not a difference in mean vehicles.

The matched distance gap is the percentage distance difference relative to the comparator, computed only on seed pairs with equal fleet size and averaged in the same way. Negative values favor the proposed configuration. It is conditional on matched fleets and is not an unconditional treatment effect.

Uncertainty comes from a percentile bootstrap (10,000 replicates, fixed seed) that resamples the five seeds and, within each size, the four groups of instances that share customer geometry, so that instances with the same coordinates

TABLE 3. Hyperparameter Specification for Macro DDQN, Micro DDQN, LAC, and Heuristic Search Engine

Component	Hyperparameter	Value / Specification
Macro DDQN	Network Architecture	Dueling MLP: Input(12) $\rightarrow$ FC(128) $\rightarrow$ LayerNorm $\rightarrow$ FC(128) $\rightarrow$ LayerNorm $\rightarrow$ Modes(7)
	Optimizer	Adam (lr $= 3 \times 10^{-4}$, Cosine Annealing to 10^{-5})
	Buffer & Batch Size	Prioritized Replay: 20,000 transitions, Batch 64
	Discount Factor (γ)	0.95
	Target Update (τ_{macro})	Polyak soft update $\tau_{\text{macro}} = 0.005$
Micro DDQN	Network Architecture	MLP: Input(20) $\rightarrow$ FC(128) $\rightarrow$ LayerNorm $\rightarrow$ ReLU $\rightarrow$ FC(128) $\rightarrow$ (Value / Adv)
	Optimizer	Adam (lr $= 3 \times 10^{-4}$, Cosine Annealing to 10^{-5})
	Buffer & Batch Size	Prioritized Replay: 30,000 steps, Batch 64
	Discount Factor (γ)	0.97
	Blending ($\alpha_{\text{p}}, \beta_{\text{b}}, \alpha_{\text{u}}, c_{\text{ucb}}$)	$0.55,\ 0.20,\ 0.35,\ 1.0$
LAC Model	Architecture	MLP: Input(9) $\rightarrow$ FC(64) $\rightarrow$ LayerNorm $\rightarrow$ ReLU $\rightarrow$ Dropout(0.1) $\rightarrow$ FC(48) $\rightarrow$ ReLU $\rightarrow$ FC(32) $\rightarrow$ ReLU $\rightarrow$ FC(1) $\rightarrow$ Sigmoid
	Lookahead Horizon (H_{lac})	80 LAC decisions
	Warmup Period	300 consulted decisions (Simulated Annealing only)
	Acceptance Threshold	Adaptive $\tau_{\text{acc}}(t) \in [0.40, 0.75]$
	Loss Function	Class-Weighted Binary Cross-Entropy
Search Engine	Granular Neighbors (k)	15 nearest customers by travel distance
	Set Partitioning (λ_{pen})	Grid: $\{2.0, 5.0, 12.0, 30.0, 50.0\}\bar{c}$, floor 200
	Route Pool Limit ($\lvert\mathcal{P}\rvert_{\max}$)	$\min(2000, 600 + 4N)$ routes
	MIP Cutoff	Per query: 3, 4.7, 8 s at $N = 100, 200, 400$ (HiGHS)

TABLE 4. Rerun at 300 s: Proposed Configuration vs. Each Comparator (Intervals at 98.33%)

Comparator	Fleet score	Seed pairs W/T/L	Matched TD gap (%)
ALNS-Base	**[PENDING:]**	**[PENDING:]**	**[PENDING:]**
PyVRP-ILS	**[PENDING:]**	**[PENDING:]**	**[PENDING:]**
NeuroLKH-adapted	**[PENDING:]**	**[PENDING:]**	**[PENDING:]**

move together. With four groups per size the intervals are coarse, and several have an endpoint at exactly zero. An interval is reported as undefined when any replicate has no matched pair in a size stratum. The three primary comparisons at 300 s use Bonferroni-adjusted 98.33% intervals, and the eight primary factorial contrasts at 300 s use 99.38% intervals. Other budgets, size strata and exploratory contrasts use unadjusted 95% intervals and are descriptive.

A missing cell caused by infrastructure failure is not an algorithm outcome, so the primary treatment drops that seed pair from that comparison only, and we report bounds that count such gaps as failures. A cell in which the algorithm itself found no solution stays in the analysis and loses its pair.

C. COMPARISON AT 300 S (RQ1, RQ4)

[PENDING: adoption rule outcome: the fleet score of F111111+GES against F111111-blind at 98.75%, and which configuration is proposed. The comparison family follows that outcome.]

[PENDING: text for Table ??, then the same comparisons by size (100, 200, 400 customers), the bounds for infrastructure gaps, the repeat without blocks started at load above 8, the two repeats based on load during a run with their block counts, and F111111+GES against F111111-blind whatever the adoption outcome.]

D. EFFECT OF THE TIME BUDGET (RQ2)

[PENDING: fleet score and matched gap at 10, 60 and 900 s, with unadjusted intervals. State whether the 900 s stage was completed. Compilation of the phase's kernels and the phase itself are charged to the budget, so state how much of the 10 s budget they used and that the 10 s comparison mostly reflects this cost.]

E. FACTORIAL ABLATION (RQ3)

The ablation uses 18 instances (the first instance of each of the six families at each size), five seeds, and budgets of 60 and 300 s. Each of six factors (GEC, SP, Macro, Micro, LAC, GNN) is set to its low level in the full configuration and to its high level in the bare configuration. Low levels are not simple removals: GEC keeps direct insertion, SP disables every recombination query, Macro and Micro use rule-based policies in place of the DDQNs, LAC uses Simulated Annealing alone, and GNN runs without a heatmap. A 2×2 design contrasts the learning bundle (Macro, Micro, LAC, GNN) with the heuristic bundle (GEC, SP). The case with all six factors low shares the search infrastructure, including the tail phases, and is not ALNS-Base. In the rerun the Guided Ejection Search phase is a fixed layer in all cases, so effects are measured on top of it.

[PENDING: table and text: the eight primary contrasts at 300 s with 99.38% intervals, and the 60 s and exploratory contrasts as descriptive.]

F. INSTANCE FAMILIES (RQ5)

[PENDING: descriptive per-family fleet and matched distance against each comparator. Say which families lose, not only which win.]

G. AVAILABILITY AND FEASIBILITY

The supervisor validates every incumbent in the original problem space, covering customer coverage, capacity and time windows, before it counts. A cell without such an incumbent is reported as unavailable, and it is never replaced by a later or uncertified result. In the first sweep, 21 of the 8,620 cells were lost to infrastructure failures. **[PENDING: number of cells without a solution by case, infrastructure gaps, blocks rerun after machine sleep, and interrupted cells in the rerun.]**

H. COMPUTATIONAL COST

[PENDING: time per component (DDQN inference and updates, GNN, ejection chains, set partitioning, certification) from a phase trace on a representative subset, and iterations per second by size, including ALNS-Base.]

I. RECORD OF THE FIRST SWEEP

The first full sweep is kept on record whichever configuration is adopted. Its hybrid read the reference fleet, had no Guided Ejection Search phase, and ran under the varying machine speed described in Section **??**, so its results are valid within blocks and do not supersede the rerun.

Table **??** shows that at 300 s this hybrid did not differ detectably from ALNS-Base or PyVRP-ILS on fleet size and used more vehicles than NeuroLKH-adapted. Where fleets matched, it used less distance than NeuroLKH-adapted on average, but that interval is undefined. Against PyVRP-ILS it was behind at 100 customers (fleet score -0.150, 95% interval $[-0.257,\ 0]$) and ahead in point estimate at 200 and 400 customers ($+0.150$ and $+0.300$), with intervals that include zero. At 10 s it was behind all three comparators on fleet size. At 900 s, on the unadjusted interval, it was ahead of ALNS-Base (fleet score $+0.135$, $[0.022,\ 0.201]$; matched gap -0.41%, $[-0.70,\ -0.03]$) and still behind NeuroLKH-adapted (-0.134, $[-0.224,\ -0.021]$).

In its factorial ablation, no primary contrast had a fleet interval excluding zero at 300 s under the 99.38% correction. On unadjusted intervals, the Micro DDQN against its rule policy in the full configuration ($+0.200$, $[0.060,\ 0.366]$) and the learning bundle added to the heuristic bundle ($+0.189$, $[0.067,\ 0.363]$) excluded zero at 60 s, and the same contrast in the bare configuration did so at 300 s ($+0.111$, $[0.022,\ 0.178]$). The evidence for learned control in this sweep is therefore descriptive.

VI. DISCUSSION AND MANAGERIAL INSIGHTS

This section interprets the results, states their practical meaning, and lists the analytical properties and limitations of the method.

A. WHERE THE METHOD DOES AND DOES NOT HELP

[PENDING: write from the rerun: what the method gains relative to ALNS-Base and at what budgets, whether it is behind PyVRP-ILS or NeuroLKH-adapted and on which sizes and families, and how much of any gain comes from Guided Ejection Search and other classical components instead of learning. Do not generalize beyond the 74 instances.]

B. MANAGERIAL IMPLICATIONS FOR COMMERCIAL LOGISTICS

In commercial freight operations, capital depreciation, vehicle financing, insurance, and driver shift wages dominate marginal fuel consumption [3], [4]. Fixed vehicle cost is therefore the main driver, and removing one vehicle saves more, and more regularly, than a marginal distance improvement. Extra compute is therefore worth spending when it removes a vehicle, but the solver that removes it need not be this one. **[PENDING: say which solver the results favor at planning budgets of 5 and 15 minutes.]**

C. ANALYTICAL PROPERTIES AND ARCHITECTURAL DESIGN PRINCIPLES

Related derivations are in Supplementary Sections 1–4, and the full algorithm is in Supplementary Section 5.

Property 1 (Lexicographic Consistency): *Under the scalar surrogate* $F(s) = \mathcal{M} \cdot N_{\mathrm{V}}(s) + D_{\mathrm{T}}(s)$ *with* $\mathcal{M} > \max_s D_{\mathrm{T}}(s) - \min_s D_{\mathrm{T}}(s)$, $F(s_1) < F(s_2) \iff s_1 \prec s_2$. With $\mathcal{M} = 10^5$ the condition holds for all evaluated instances, whose travel distances are below 10^5; the property concerns the comparison of evaluated solutions and implies no optimality guarantee.

Property 2 (Granular Candidate Space Reduction): *Restricting each customer to its* $k = 15$ *nearest customers reduces the candidate arc space from* $N(N-1)$ *to* kN *arcs, a reduction of* $\Gamma_{prune}(N) = 1 - k/(N-1)$. This removes 84.85%, 92.46% and 96.24% of customer arcs at $N = 100$, 200 and 400. This is arithmetic on the candidate lists and is separate from the GNN thresholds in Section IV-E3.

Design Principle 1 (Multi-Scale Penalty Grid): *In Set Partitioning over Route Pool* $\mathcal{P}$*, the penalty schedule* $\lambda_{pen} \in \{2.0, \ldots, 50.0\}\bar{c}$ *shifts the model from preserving distance to removing vehicles.* At $\lambda_{\text{pen}} = 50.0$, the vehicle penalty strictly dominates partition distance variance, enforcing fleet minimization during HiGHS solves [32].

Design Principle 2 (Downstream Lookahead via LAC): *Unlike Simulated Annealing, where non-improving acceptance vanishes (*$P_{SA} \to 0$ *as* $T \to 0$*), LAC evaluates structural transition promise using a delayed observational label (*$H_{lac} = 80$ *LAC decisions).* Retaining moves with predicted probability $p_t \geq \tau_{\text{acc}}$ is intended to aid plateau escape; this is an empirical design choice, not a convergence guarantee (Section IV-D).

D. LIMITATIONS AND THREATS TO VALIDITY

The main limitations are these. (i) Benchmark Scope: the evaluation covers 74 instances with four geometry groups per size and few large instances, so it says little about other distributions. (ii) MIP Dependency: Set Partitioning relies on HiGHS throughput within a per-query time limit. (iii) Compute Overhead: online neural updates and Set Partitioning consume part of each wall-clock budget that would otherwise go to search iterations. (iv) Static Scope: the framework targets static VRPTW. (v) GNN Scale Shift: the GNN was trained and validated with one training seed at $N = 100$ and 200 only, so use at $N = 400$ is an extrapolation, and its validation recall is measured against heuristic labels. (vi) LAC Disentanglement: because LAC operates disjunctively (Accept $\iff p_t \geq \tau_{\text{acc}} \vee P_{\text{SA}} > U$), future work is needed to isolate predictive lookahead from generic acceptance volume. (vii) Effective Compute: machine speed varied between blocks of cells (Section **??**), so results are comparable within a block and do not describe a fixed amount of compute. (viii) One Machine and Fixed Baseline Settings: all runs used one notebook, and the baselines were not tuned for these budgets. (ix) Test-Informed Development: test results motivated the Guided Ejection Search phase, and the development

TABLE 5. First Sweep at 300 s: F111111, Which Read the Reference Fleet, vs. Each Comparator (Intervals at 98.33%)

Comparator	Fleet score	Seed pairs W/T/L	Instances W/T/L	Matched TD gap (%)
ALNS-Base	$+0.076$ [-0.010, 0.134]	38/320/10	16/54/4	-0.23 [-0.65, 0.11]
PyVRP-ILS	-0.065 [-0.183, 0.083]	35/275/59	8/48/18	$+0.84$ (undefined)
NeuroLKH-adapted	-0.176 [-0.284, -0.029]	8/286/73	2/50/22	-4.62 (undefined)

Instances W/T/L compares mean fleet over five seeds and is descriptive. Treating infrastructure gaps as failures changes no conclusion.

instances share coordinates with test instances (Section **??**). Every incumbent counted in the results is validated in the original problem space by the supervisor before it counts, and cells without a valid incumbent are reported as unavailable.

VII. CONCLUSION

This paper presented Tri-Level Hybrid DDQN-ALNS, a hierarchical learning-augmented method for the VRPTW. It separates macro plateau control (Dueling DDQN SMDP) from micro operator selection (Dueling DDQN with Softmax-Entropy policy gating) and predictive lookahead acceptance (LAC). Feasibility rests on insertion checks against cached route profiles during search and on validation in the original problem space before any solution counts.

[PENDING: write from the rerun at 300 s, including losses and the ablation result.] The method could be extended to electric vehicle routing with non-linear charging and to real-time stochastic dispatch. All benchmark instances, trained checkpoints, baseline implementations, evaluation scripts, and raw execution logs across all 74 instances $\times$ 5 seeds are available at https://github.com/Thundercok/vrptw-neural-hybrid-optimizer.

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